

A Universe in a Bottle — The Eleven-Paper Series

A peer-review-style write-up of the BCC quantum-cellular-automaton (QCA) model of physics. Each paper is self-contained and replication-grade: it states the construction, gives the derivations, quotes the verification residuals, names the underlying findings (`findings/F*.md`) and modules (`ca-simulation/ca_*.py`), and lists external references.

Issued 2026-06-08. Supersedes the ten-paper draft in `deprecated/papers-v1/` (the matter sector is now split into three papers — fermions, leptons, baryons — and the dielectric black hole has its own paper). Canonical SI cell aligned to Finding F107 throughout: $a = \sqrt{8\pi} 3^{1/4} \ell_P = 6.5978 \ell_P = 1.0664 \times 10^{-34} \text{ m}$.

*Revision (2026-07-01): the horizon-free “dielectric black hole” paper (formerly Paper VIII, F114) is **deprecated** following the adoption of the full-tensor induced Einstein equation (F178; see Paper VII §4.2) — the exact strong-field object is Schwarzschild/Kerr, and the dielectric is its vacuum/weak-field representation. The deprecated paper is retained at `deprecated/Paper-08-Dielectric-Black-Hole.md`. Papers IX–XIII are renumbered consecutively to VIII–XII.*

Reading order

#	Paper	What it establishes
I	Base Structure	BCC Weyl QCA substrate; speed of light as a rotation rate ($c_{\text{lat}} = 1/\sqrt{3}$); Maxwell as a linearised rotation; the spherical-Pythagorean mass shell.
II	The Photon	The photon as a bound pair of two spin- $\frac{1}{2}$ lattice quanta; massless, luminal, transverse, exactly non-birefringent; supersedes the chiral -bilinear.
III	Higgs-Free Mass	Chiral- $SU(2)$ complex-mass step (no Higgs); the $SU(2)_L$ Ward identity; the NJL constituent-mass scale; the Koide relation as cubic equipartition.
IV	Electromagnetism	Maxwell from the rotation rule; $U(1)$ minimal coupling forces the paired photon; Aharonov–Bohm + sourced curl end-to-end; charge quantisation and anomaly freedom.

#	Paper	What it establishes
V	Strong Force	Dynamical $SU(3)$ gluons (even propagator, forced); confinement from exact 2D area law $\rightarrow$ colour-dielectric dual superconductor $\rightarrow$ gauge Monte-Carlo; centre-phase closure.
VI	Weak Force	Chiral $SU(2)_L \times U(1)_Y$ without a Higgs; rank-1 Stueckelberg masses; the derived Weinberg angle ($\sin^2 \theta_W = \frac{1}{4}$); end-to-end β -decay.
VII	Gravity	Gravity as an impedance-matched lattice dielectric ($K = e^{2GM/rc^2}$, PPN $\beta = \gamma = 1$), reclassified as the vacuum/weak-field representation of the canonical induced Einstein equation; the structural Newton constant $G = a^2 c^3 / (8\pi \sqrt{3} \hbar)$; exact Schwarzschild/Kerr in the strong field.
VIII	Fermion Sector	Weyl/Dirac quanta; chirality-helicity correspondence; exactly three generations from O_h (no fourth); anomaly-free first generation.
IX	Lepton Sector	Higgs-free hypercharge; the Koide 45° equipartition as an EM-selected critical point; the Higgs-free see-saw neutrino.
X	Baryon Sector	The colour-singlet proton; centre-phase closure (baryon mass = field energy); the dynamical pion; the deuteron bound by the tensor force; the derived NN repulsive core.

#	Paper	What it establishes
XI	One Currency (Unification)	Synthesis: mass, energy, light, and gravity are one rotation rate Ω read four ways; $E = mc^2$ makes the energy-coupled dielectric gravity <i>forced</i> , not merely permitted. (Draws on F26/F167/F69/F68/F168/F64/F106.)
XII	The Koide Angle	The charged-lepton phase : $\delta^* = \frac{2}{9}$ as the E_g representation weight $\dim E_g / \dim(T_{1u} \otimes T_{1u})$; unified with the Koide amplitude by the saturation self-duality $3\delta = Q$ (radial half BPS-derived, angular half = the one shared residual); full charged-lepton shape to $\leq 0.007\%$, zero shape parameters. (Findings F174–F177.)

Headline verified numbers

- $c_{\text{lat}} = 1/\sqrt{3} = 0.5773503$ (rotation-rate slope, axis and body diagonal)
- $\sin^2 \theta_W = \frac{1}{4} \Rightarrow m_Z/m_W = 2/\sqrt{3} = 1.1547$ (zero-parameter; 1.77% from PDG)
- Koide $Q = 0.6666605$ (0.91σ from $\frac{2}{3}$); $m_\tau^{\text{pred}} = 1776.97$ MeV (6.1×10^{-5})
- $a/\ell_P = \sqrt{8\pi} 3^{1/4} = 6.5978$; $G = 6.6743 \times 10^{-11}$ SI (3×10^{-8} , CODATA round-off)
- Strong-field black hole: exact Schwarzschild/Kerr (horizon, shadow $3\sqrt{3}M$); the earlier horizon-free dielectric-metric shadow prediction ($2e/3\sqrt{3} = 1.0463$, +4.63%) is superseded (deprecated, F114→F178→F183)
- Deuteron $E_b = 2.224$ MeV, $\kappa = 0.2316$ fm $^{-1}$; baryon constituent-mass fraction 0.96% of m_p

Standing design decisions (deviations from the Standard Model)

1. Speed of light = rotation rate, not phase velocity (F26).
2. Mass from chiral- $SU(2)$ complex coupling, not a Higgs field; hypercharge on the same gauge field (F27/F41).
3. Photon = bound pair of two spin- $\frac{1}{2}$ quanta, not a fundamental spin-1 boson (F69).
4. Gravity = the induced Einstein equation sourced by the full stress-energy tensor (F178); the single impedance-matched lattice dielectric $K = e^{2GM/rc^2}$ (F64) is its vacuum/weak-field representation.

See docs/theory/key-decisions.md and findings/ for the full record.